NASM VIRTUAL COACHING SPECIALIST (VCS)

Chapter 3: Technology Considerations

National Academy of Sports Medicine — Evidence-Based Telehealth Curriculum

Chapter 3Technology Considerations
Lesson 1Hardware Requirements

Virtual training relies heavily on technology, which, when utilized properly, can improve training efficiency and outcomes, facilitate better communication and client relationships, and improve the ability to effectively manage more clients. From filming high-quality videos to choosing the correct video conferencing software, this chapter will review the necessary and ancillary technologies that will help fitness professionals develop, run, and scale a successful virtual training business.

Learning Objectives

After completing this chapter, the NASM Virtual Coach will have the ability to do the following:

- **Determine** hardware, software, and operating system requirements needed to operate a virtual coaching business.
- **Select** an appropriate internet service and computer peripherals to suit the needs of a virtual coaching business.
- **Identify** fitness programming software to help manage and assess client progress.

Introduction

Fitness professionals have an opportunity to use technology to develop, operate, and scale successful virtual training businesses. Clients are increasingly utilizing non-traditional, technologically driven fitness solutions, including mobile workout and food logging apps, wearable fitness technology, and virtual training programming. Accordingly, fitness professionals who are proficient in computer, mobile, and wearable technology can offer viable fitness solutions and meet the rapidly evolving needs of consumers within the digital market. This digital market expands the reach of a fitness professional from what was once just a local impact to now a global one.

Within the fitness industry, fitness professionals might find technology to be a major disruptor to the traditional approaches and methods in the delivery of services and client management. However, as technology continues to advance, fitness professionals have an opportunity to adopt new approaches to engage clients, deliver a quality service, and compete effectively in a demanding market. This chapter will explore the technological infrastructure, including hardware, software, peripherals, Internet service requirements, and wearable fitness technology considerations that are necessary to operate a successful virtual training business.



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Technology Considerations Checklist

A Technology Considerations Checklist is provided for the fitness professional to use throughout the chapter to track technology considerations, notate inventory the fitness professional may already own, and to suggest future technology purchases.

Technology Considerations Checklist

Hardware Requirements

Hardware includes the physical components that are required for a computer system to function. These physical components include a mouse or trackpad, keyboard, screen or monitor, and speakers. In addition, the physical components of processing hardware, which can be thought of like puzzle pieces (**Figure: Physical Components of Processing Hardware**), include the following:



FigurePhysical Components of Processing Hardware

Computer Processing

The first puzzle piece of a computer's hardware requirements is computer processing. When choosing a computer, it is important to ensure the processing capability aligns with the intended use of the machine. A personal home computer used for basic computing, some streaming, and Internet access will have significantly different processing needs than a computer intended for small business use. In order to make the best purchase for building a virtual coaching business, the fitness professional should understand what components and requirements are needed to operate effectively, efficiently, and productively.

The **central processing unit**, or CPU, is the brain of a computer (**Figure: CPU Chip**). It carries out and directs all tasks being performed. The faster a CPU is, the faster and more efficiently the computer will run. A fitness professional may be using Excel spreadsheets for exercise programming and have a video being uploaded to their workout directory for clients, while also using instant communication applications on their computer—all of which can consume processing power.



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FigureCPU Chip

Having enough processing capacity to keep a computer running at optimal pace, while multiple programs are running, can ensure productive workflow.

CPU speed is currently measured in **gigahertz** (GHz), but, historically, when processing power was lower, they were in megahertz (MHz). A gigahertz is also known as "Clock Speed." At 2.5 GHz, it is like a clock that *ticks* 2.5 billion times in a second. Each *tick* is an opportunity for the CPU to move and modify the smallest fragments of data called **bits**. The higher the GHz classification for a processor, the quicker a processor can process data and make decisions. Modern computers typically range from 2.0 GHz through to 5 GHz or more as technology advances. While top-of-the-line processor speeds are not needed for virtual coaching, the fitness professional should ensure the machine being purchased has between 2.4 and 4.0 GHz of processing speed.

Getting Technical

Another component of CPU technology to consider is how many cores the processor has. Each core can work on a single task. The more cores a processor has, the more tasks a CPU can process. An average computer may come with a dual core processor but can contain more. Ideally, aiming to purchase a

machine with four cores or more will help with multitasking and running multiple programs without slowing the computer down.

Central processing unit (CPU)

A CPU is the brain of a computer. It carries out and directs all tasks being performed.

Gigahertz

The measurement of how fast a computer processor can manage bits of data – a 1.0 GHz processor can process 1 million bits of data per second; a 2.4 GHz processor can process 2.4 million bits of data per second.

Bits

The smallest fragments of computer data.

RAM

RAM stands for Random Access Memory and is another piece of the puzzle that makes up a computer system (**Figure: RAM**). RAM is essentially a computer's short-term memory.



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FigureRAM

RAM temporarily stores the information that the computer needs to access immediately or in the near future.

RAM keeps track of everything that is being processed by the computer such as open web browser tabs, a training spreadsheet being edited, or a video being rendered. Instead of the computer going into a hard drive storage device to pull the data it requires, RAM knows what the computer will need next and keeps that information at hand, ready to be used.

For example, RAM is like having a grocery shopping list in hand at the store. Having this list shortens the shopping trip; without it, the shopper would walk aimlessly up and down the aisle trying to remember what is needed for groceries and extend the time spent shopping!

Getting Technical

RAM is short-term memory, whereas a hard drive is used for long-term storage. In fact, RAM does not actually *store* anything. RAM is what is considered volatile memory, meaning it loses its data as soon as the power is shut off. The data and information are only housed temporarily, as needed. Hard drives are considered non-volatile memory, meaning the data is stored even when the power is turned off.

RAM

Random Access Memory – A computer's short-term memory for processing immediate and upcoming tasks.

How Much RAM is Needed?

RAM is currently measured in Gigabytes (GB). How many GB of RAM a computer has will determine how much data it can temporarily store for instant access as the computer is operating. The amount of RAM needed will depend mainly on what the computer is being used for and the complexity of the program. The more complicated a program becomes the more RAM will be required. A simple program, such as a basic notepad, will not take up much RAM, if any. Complex programs like Excel, Photoshop, or a video-editing software may use gigabytes of RAM by themselves.

With virtual coaching requiring programs like Excel spreadsheets, video editing software, or video conferencing programs that can consume a large amount of RAM by themselves, it is a good idea to ensure you have enough RAM rather than not enough (**Table: RAM Recommendations**). This can also minimize expensive RAM add-on kits or having to purchase a new computer if there is no option to expand RAM capabilities. It saves money in the long-term and minimizes frustrations with a slow operating computer system.

RAM Recommendations

<4 GB	Minimum requirements for operating systems and basic function.
4 GB to 8 GB	Minimum requirements for typical computer use.
8 GB to 16 GB	Provides enough RAM to support most or all tasks except high-end video editing or game rendering.
16 GB or more	Professional-grade editing software, gaming, and content rendering.

Coach’s Corner

In most cases, 8 to 16 GB of RAM will be enough to effectively and efficiently run a virtual coaching business. When purchasing a laptop or desktop, ensuring it has a minimum of 8 GB of RAM, is a great starting point. 16 GB of RAM would be enough for the majority of the process and software requirements for running a virtual coaching business.

Food for Thought

More RAM is not always better. While it is a good idea to get more than needed, rather than not enough, getting 32 GB of RAM when the max operating requirements are less than 16 GB may be wasteful. It costs more money and does not make the computer faster unless the software programs actually need 16 GB+ of RAM.

Storage

A third piece of the computer hardware requirements puzzle is storage. With virtual coaching, the fitness professional is responsible for not only handling and organizing client information, but also managing and storing business documents such as a business plan, workout demonstration videos, and any of created social media content. From liability and sign up forms, to progress photos, workout plans, and private client information, being able to safely store this data as well as quickly access it as needed is key to running a successful virtual coaching business. There are two main options for data storage (**Table: Types of Storage**):

1. Local Data Storage
2. Cloud or Online Storage

Types of Storage

	Local Data Storage	**Cloud or Online Storage**
What it is	Hard copy of the information or data Examples include the following: • USB flash drives • Internal hard drives • External hard drives	Information stored on the Internet, on a server Examples include the following: • Google Drive • Dropbox • iCloud
Accessibility	• Harder to access on the go, must carry a physical hard drive	• Easy to access on the go from smartphone or laptop
Security	• Increased risk of physical drives being stolen, lost, or damaged • Added security because the fitness professional controls who has access to storage devices	• No risk of data loss due to mishandling, damage, or loss of a device • Increased risk of data-hacking or online data theft due to third-party servers hosting data

Scalability	• Expensive to scale due to needing to use physical hard drives • Physical hard drives take up more physical space	• Added cloud storage costs minimal compared to physical hard drives
Speed of Use	• Faster than downloading or uploading files from cloud-based storage	• Large files can be time consuming to upload or download

Local Storage

Local storage has the benefit of instant access and is typically faster than uploading and downloading information from the cloud. Plus, if an Internet connection goes offline temporarily, a virtual coaching business can still operate because of the data being accessible on demand, as needed. Local storage devices such as thumb drives, the physical storage device on the computer, or an external drive (**Figure: Local Storage**) are also more secure than cloud storage because information can only be physically accessed with those storage devices in hand. However, if those drives become lost, stolen, or crash, there is a risk of data loss.

There are two main types of hard drives to consider for local storage:

- **Hard Disk Drive or HDD:** A physical disk that spins inside a case where the data is written and erased. They are typically cheaper but also larger and slower than other types of hard drives.
- **Solid-State Drive or SSD:** A solid-state storage device is similar to a thumb drive. It has no moving parts and is faster than a standard HDD but also comes at a higher cost.



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FigureLocal Storage
Food for Thought

Consider getting an external hard drive that is kept at home, which regularly backs up all the vital information for running a virtual coaching business. If a computer or phone is lost, or something gets damaged, that external drive will safely store all that data and it is easily accessible.

Hard disk drive (HDD)

A data-storage device, which has a physical spinning disk where information or data is stored, written, or deleted.

Solid-state drive (SSD)

A data-storage device that uses no physical spinning or moving parts to store, write, or delete data.

Cloud Storage



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Cloud storage has the advantage of being accessed on the go with most cloud services being accessible from a phone or laptop. This means a virtual coaching business can be easily conducted from a smartphone or laptop and a Wi-Fi signal. However, there may be higher security risks with cloud-based storage. Because the data is stored in a third-party system, there may be some added risk of attackers hacking into the system and gaining access to the data (Tujilio et al., 2016). Alongside added security risks, large file uploads and downloads may be more time consuming than transferring between a computer and a physical external hard drive.

When picking a cloud storage service, be mindful of scaling a virtual coaching business. Dropbox and Google Drive both come with word processing and spreadsheet capabilities and a large amount of initial free storage. Google Drive has some added benefits such as integrated e-mail and business services and other professional software such as Google Sheets (spreadsheets), Google Docs (word processing software), and the ability to create forms for collecting client information and sign ups.

Both local storage and cloud storage have their pros and cons, and using a mixed approach of both cloud storage and local storage will be most effective for managing and running a virtual coaching business.

Video Card

The final piece, a **video card**, or graphics card, is hardware that takes all the data from across a computer system and generates it into a visible image on the computer monitor or laptop screen (**Figure: Video Card**). Most video cards include a stand-alone processing unit, built-in memory that is similar to RAM, and sometimes a cooling fan to help manage temperatures. It is essentially another computer in itself. Just like RAM, a video card is designed to take added stress off the CPU to more efficiently process images and transfer them to the monitor or screen.



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FigureVideo Card

Many graphics cards will also have additional outputs such as an HDMI connection or VGA connection. HDMI connections transmit both audio and video data, whereas a VGA connection only transmits video data. Some graphics cards will also include an S-video display and USB port, which can connect to digital cameras, webcams, and other peripherals like microphones.

For doing high-quality exercise videos, filming and editing social media content, or creating PowerPoint presentations, the built-in video card that comes with any mid-range desktop or laptop will meet the needs of a virtual coaching business. Upgrading video cards might be considered if the machine being purchased will be used for high-definition video editing, gaming, or other professional editing software uses.

Video card

A video card or graphics card is a piece of hardware that takes all the data from across a computer system and generates it into a visible image on the computer monitor or laptop screen.

Desktop or Laptop?

A fitness professional should consider whether a desktop computer or laptop computer will be needed to build a virtual coaching business. There is more available processing power in a typical desktop computer than in a laptop computer when looking at a comparable price range. While a laptop has all of the hardware requirements built into it as a complete package, a desktop computer will have an added cost of the wires, monitor, speakers, mouse, and keyboard. Once these pieces are factored in, a high-quality laptop can be purchased for a similar price with more than enough processing capabilities for a virtual coaching business.

If building a virtual coaching business with a home base in mind and mobility is not a primary consideration, a desktop computer will give the most capability to the end user. However, the mobility of a laptop often outweighs the benefits of a desktop computer when it comes to running a virtual coaching business. A laptop can be used in a variety of spaces such as a coffee shop, throughout a home, and when travelling. All that is needed is a stable Internet connection.



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Software Requirements

Software components are the actual programs running on a computer system to make it function and operate. Software is strings of programming code that are run and processed by the hardware components of the computer system. For example, software is found in the form of operating systems themselves such as Windows Operating System from Microsoft or iOS from Apple, to programs such as the Word processor, Excel, video games, and even just music players. This topic will cover operating systems, video conferencing systems, and other software recommendations to help the fitness professional operate in the virtual coaching space effectively and efficiently.

Operating Software

There are currently three major operating systems dominating the market in computer technology. Microsoft with their Windows Operating System, Apple with their iOS and OS-X operating systems, and Linux, which is used as a baseline in many systems like Roku TVs, social media feeds, and smart technology integrated into the automotive industry. In 2019, Microsoft dominated the software industry with a worldwide market share of 77% with Windows and Apple Inc. with OS-X around 17% (Statcounter GlobalStats, 2020). Both Microsoft and Apple operating systems have their own unique pros and cons.

The primary consideration for purchasing a computer system and for which operating system to use will always be governed by the fitness professional's budget. Apple products have always carried a higher price tag when compared to similar performing Windows-based computers.



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Food for Thought

The 2019 baseline MacBook Air was priced at $1,199 USD in 2019. A similarly performing Windows-based laptop was priced around $200 cheaper. Higher-performing Apple computers can be even more expensive, ranging upward of $2,500 dollars.

Apple operating systems do come with some extra software and programming that may need to be purchased on a Windows-based operating system. With iMovie, which can be used to edit videos, merge sound clips, and add subtitles, for example, purchasing third-party video editing software may not be needed. However, Windows third-party software compatibility shadows the Apple operating system.

The notable difference between the Windows and Apple operating systems will be in the user experience and interface. For example, the OS-X system from Apple uses a pop-up app dock for favorite apps, whereas Windows OS has a taskbar and a start menu. Windows has a more versatile file management system, whereas OS-X offers a higher degree of integration with other Apple products. While the brands themselves and loyal fans may proclaim that one operating system is better, preference becomes the main determining factor of how to choose which OS to use.

Overall, the main determining factors when choosing an operating system to fit the needs of a virtual coaching business should be the following:

- Budget
- Intended use, such as added video editing
- Ease of use, with peripherals like tablets or wearable fitness trackers
- Preference, user experience, and interface
- Hardware

Critical!

The fitness professional does not need to purchase a new computer or components. Purchasing a preowned computer can save money when on a tighter budget.

Video Conferencing Software



FigureVisual Connections

Video conferencing technology is a critical channel of communication for virtual training. Not only does video conferencing eliminate the barriers of geographical location, it allows fitness professionals to engage and connect with clients in a more meaningful manner.

In a study on processing information, humans process visual information more quickly and capably than through audible or written information (Cohen et al., 2009). This transfers to improved engagement and also client understanding, which are crucial components of successful client outcomes. Regular eye contact throughout video calls may result in higher perceived friendliness and improved information recall. A pleasant, smiling facial expression can indicate openness and positively strengthen the interaction between the fitness professional and the client (Park & Park, 2018) (**Figure: Visual Connections**).

In addition, studies have shown that body language and tone of voice contribute to 55% and 38%, respectively, of effective communication; while words, such as a phone call or written message, only account for around 7% (Park & Park, 2018). Video conferencing allows the fitness professional, and the client, to convey emotion and tonality in a conversation or virtual training session that may otherwise get lost through other forms of communication. Some examples might include the following:

- The fitness professional can sense if the client has encountered some obstacles if the tone of the conversation shifts.
- The fitness professional can share in a client's excitement if they hit a personal record in one of their exercises.
- The fitness professional can explain and demonstrate proper exercise form.
- When having difficult conversations, the client can hear the fitness professional and see if they are paying full attention, helping the client find solutions.

There is no shortage of options when it comes to video conferencing. When choosing a video conferencing software, the user should consider factors such as price, participant and meeting time thresholds, mobile app accompaniment, and other features such as meeting recording. While many video conferencing applications are available for free, it is not uncommon for them to have restrictions on the number of attendees and meeting runtimes. Several software vendors offer the user the ability to upgrade to a paid version, which removes the aforementioned restrictions. The following table outlines some of the more popular options currently available; however, these metrics are subject to change (**Table: Comparing Video Conferencing Software**). Before choosing a video conferencing service, investigate if the company has changed their terms and conditions.

Comparing Video Conferencing Software

Software	Paid/Free	Maximum No. of Participants	Maximum Meeting Runtime	Mobile App	Includes Recording Cloud Storage?
Zoom	Free	100	40 min	Y	N
Zoom	Paid	100+	24 hrs	Y	Up to 4 Gb
Web-Ex	Free	100	50 min	Y	N
Web-Ex	Paid	200+	24 hrs	Y	Up to 10 Gb

Google Meet	Free	150	60 min	Y	N
Google Meet	Paid	100+	300 hrs	Y	Y, Varies
Microsoft Teams	Free	50	24 hrs	Y	N
Microsoft Teams	Paid	250+	24 hrs	Y	Varies, paid add-on

Word Processing Software

Word processing software is another common communication tool used in virtual coaching. Word processing software is essential to the development of coaching plans and other personalized content intended for sharing with a client. It can also be used for record keeping and business administration activities such as scheduling and invoicing. Most computers are equipped with a trial version word processing software as a standard feature.

The standard software will vary depending on the operating system. For example, the Windows operating system is commonly equipped with Microsoft Word, whereas the Apple OS-X operating system is commonly equipped with Pages. Most trial version word processing software is restrictive in the features necessary to create, edit, save, and share content, which means the fitness professional may need to purchase additional software.

In addition to standard equipped software or purchased software, a user may choose to use a cloud-based word processing program. While standard equipped word processing software does not vary remarkably between programs, there are considerable differences between a cloud-based and non-cloud-based word processing software, specifically in the ability to collaborate in real time.

When choosing a word processing software, the user should consider cost, compatibility with the operating system, and features. Cloud-based versions of commonly recognized software, such as Microsoft Word or Google Docs, are often seen as advantageous due to the collaboration functionality. This allows multiple users to view and edit documents in real time with 24/7 access by both client and coach and saved version history. This functionality could be viewed as beneficial and more efficient in providing client feedback than a non-cloud-based software.

Spreadsheets

The fitness professional would benefit from spreadsheet software. This software can allow the fitness professional to create workout programs, invoice clients, or build client metric monitoring sheets, among many other uses (**Figure: Example Spreadsheet**). Google Sheets, Numbers from Apple Inc., or Excel from Microsoft are all examples of spreadsheet programs.

This program, alongside a word processing application, may be all a fitness professional needs to operate their business. One consideration, however, if business begins to scale, is that there are apps and coaching software programs that can improve time management and may be easier for the client to use.



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FigureExample Spreadsheet

Internet Service Requirements

The nature of virtual coaching relies on the ability to access the Internet with a reliable connection. Slow or inconsistent connections lead to frustrations for both the client and the fitness professional and can impact quality of service. With many Internet service providers available and variables in delivery of Internet service (fiber optic, cable, or satellite), it can be difficult to choose what is most effective for running a reliable virtual coaching business. The largest determining factor will be what is available in the area where the virtual coaching business is located and how it operates. If the fitness professional lives in a rural area, there might be limits to Internet services such as satellite or dial-up. If the fitness professional is in an urban area, likely there will be access to high-speed cable or fiber optic Internet services with high rates of bandwidth.

Bandwidth is the maximum data transfer rate of a network or Internet connection, in a specific amount of time, or how fast a person can download or upload information to and from the Internet. Bandwidth is measured in megabytes per second or Mbps. A **Megabyte** (Mb) is broken up into smaller units known as **Kilobytes** (kbit) and there are 1,000 Kilobytes in each Megabyte of data. Kilobytes are broken up into bits and there are 1000 bits per kilobyte. There is also a **Gigabyte**, which is 1,000 megabytes (Mb).

Bandwidth

The maximum data transfer rate of a network or Internet connection, in a specific amount of time, or how fast Information can be downloaded or uploaded to and from the Internet.

Megabyte

A measurement of data, 1,000 kilobytes.

Kilobyte

A measurement of data, 1,000 bits.

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There are some basic considerations for how much bandwidth is needed to operate a virtual coaching business. If providing online group training via a video conferencing platform where most of the data transfer comes from, slower Internet speeds may be utilized without an effect on the user experience. For example, Zoom, Skype, and Google Meet require as little as 1.5 Mbps for voice or low-definition video calls and up to 8 Mbps for large group video conferencing or HD video calling. A 25 Mbps Internet speed may be enough in this case.

If creating video content, performing large live stream feeds, or regularly downloading and uploading large files is integral to how the virtual coaching business is run, getting a higher bandwidth connection can increase productivity. For live-feed video conferencing, a slower Internet speed may impact the video feed quality and end-user experience (**Table: Comparing Bandwidth Speed and Download Times**).

Comparing Bandwidth Speed and Download Times

Length and Type of Media	Average Size	1 Mbps Connection Speed	5 Mbps Connection Speed	10 Mbps Connection Speed	20 Mbps Connection Speed	100 Mbps Connection Speed	1,000 Mbps Connection Speed
4-min song	4 Mb	30 s	5 s	3 s	1.5 s	0.3 s	0.03 s
10-minute client catch-up video call	30 Mb	3 m	40 s	26 s	13 s	2.5 s	0.2 s
9-hr audiobook	110 Mb	10 m	2 m	1.5 m	46 s	9.2 s	0.9 s
60-min Zoom live workout	200 Mb	20 m	5 m	3 m	1.5 m	16 s	1.7 s
45-min TV show	600 Mb	1 h	15 m	8.5 m	4 m	50 s	5 s

2-hr movie	1.0 to 1.5 Gb	2 h	24 m	21.6 m	10.5 m	1.5 m	8 s
10-minute HD 1080p workout video	4.0 Gb	6 h	72 m	60 m	32 m	4.4 m	25 s

Dial-up will have the slowest rates of bandwidth at just 56 kbits or 0.056 Mbps. When compared to the U.S. average broadband speed of around 135 Mbps, dial up is over 2,000 times slower. Just 0.3% of the U.S. is still using dial-up Internet. This is just enough to do basic tasks like answer e-mails and read webpages on the Internet at a slow pace. While dial-up is still available in some rural places, this Internet connection would not be suitable for running a virtual coaching business.

Satellite Internet is used by 6.9% of the U.S. population (Clement, 2019) and has a limitation of around 100 Mbps, with most services capping around 25Mbps. While this is enough to run a virtual coaching business, satellite Internet may be more expensive and less reliable than a cable or fiber Internet connection. Satellite Internet also operates on the *line-of-sight* principle. Meaning the satellites must be able to physically see each other to form a connection. A simple storm with poor weather conditions can knock out the Internet connection due to the satellites not being able to connect. This can make running a business a hassle if the connection to the Internet is unreliable. Additionally, satellite Internet tends to be more expensive.

Broadband Internet connections such as DSL, fiber optics, and cable Internet will be the fastest, most reliable, and often cheapest Internet connections to choose from. With bandwidth speeds from 25 Mbps to 1,000+ Mbps and reliable connections that do not rely on line-of-sight, a virtual coaching business can operate efficiently and reliably.

In order to be effective, efficient, and productive, consider signing up with an Internet service provider that can provide 25 to 75 Mbps of bandwidth. This will ensure smooth operation when uploading training videos, administering virtual coaching sessions, or performing video consultations.

Try This

Try searching for “Internet Speed Test,” and then test your current Internet speed. If it is running less than 25 Mbps, it may be time for you to upgrade your Internet package or Internet service provider.

Computer Peripheral Requirements



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By now, the fitness professional should be able to determine what hardware and software requirements are needed, how much bandwidth is needed to run a business, and what operating systems are preferred when purchasing a computer. One last consideration is computer peripherals, which may help with producing and delivering quality content to the client. The fitness professional should remember that these additional peripherals are not requirements but, rather, components that contribute to producing and delivering higher-quality content. A fitness professional may be able to start and operate a virtual coaching business solely using a smartphone and then upgrade as their budget allows.

Computer peripherals include the following:

- Lighting
- Cameras
- Microphones

Lighting

A major consideration for creating high-quality, deliverable digital content is lighting. Lighting may impact overall content quality, considering that not all training facilities or office spaces will have adequate lighting. A lack of bright natural light when filming technique videos, live workouts, or virtual coaching sessions can be a major distraction. The fitness professional should mitigate this with a well-lit environment.

The main factors to consider when purchasing lighting peripherals are:

- Lighting Spectrum
- Budget and Portability

Lighting Spectrum

Lighting spectrum or color temperature is what hue of color is being provided by a light source. Degrees of Kelvin (K) is how the appearance of light is measured. Kelvin is measured from 1,000 K up to 10,000 K. The lower end of the Kelvin spectrum has a far-red tinge to it. The high end of the Kelvin spectrum has a blue overtone.

The typical lighting spectrums used in video content is between 3,200 K and 5,500 K. The 3,200 K lighting spectrum is typically associated with indoor lighting and indoor scenes; it will be warmer in color with a higher degree of orange or yellow. The 5,500 K spectrum is a cooler light that is mostly bluish-white and is representative of midday daylight. Too low of a lighting spectrum will produce more flat colors and may create unnatural skin tone, while too high of a lighting spectrum will lack warmth and be harsh to look at.

The 5,600 K spectrum may be more optimal for use where there is a lower level of light or in indoor settings where incandescent bulbs or soft, warm lighting may be used. The midday lighting spectrum will help produce a brighter, more natural and balanced feel. The warmer 3,200 K spectrum may be more optimal for use where there is harsh white light such as under fluorescent bulbs or when filming in brightly lit locations. The warmer spectrum may reduce the harsh look that higher spectrum lighting may produce.

Many of the available lighting options will have adjustability from 3,200 K to 5,600 K, which means the fitness professional will be able to adjust the lighting spectrum accordingly for where they will be filming the majority of their content. Having this adjustability can minimize purchasing extra equipment or lighting options.

Table: Light Source Examples and Color Temperature compares light sources and their color temperature, whereas **Figure: Color Temperature in Kelvin** shows a visual representation of the color change through the Kelvin (K) scale.

Light Source Examples and Color Temperature

Light Source Example	Color Temperature in Kelvin (K)
Blue Sky	10,000 K
Shade on Sunny Day	8,000 K
Overcast Sky	6,000 K
Natural Daylight	5,500 K
Incandescent Bulb	3,400 K
Sunrise and Sunset	3,200 K
Candle	1,500 K



FigureColor Temperature in Kelvin

Getting Technical

Lighting spectrum also interacts with camera settings and can help improve the end product. There are several online resources available that provide in-depth discussions and tutorials regarding this topic.

Budget and Portability

Producing high-quality content that is valuable to the client does not require purchasing high-end equipment. While there is studio quality lighting available at a steep price, a fitness professional just starting out who is looking to film training videos and perform virtual coaching sessions will not need such high-end equipment. Building a virtual coaching business also means building a business that is portable. With much of the technology to run a business being available on a smartphone, tablet, or laptop, working on the go will present some added challenges if it means having to carry around bulky equipment.

With budget and portability in mind, the fitness professional may find appropriate solutions in portable ring lights or clip-on smartphone lights that are both compact and effective in meeting lighting requirements and lighting spectrum applications. The ring lights are typically adjustable (**Figure: Ring Light**), while the clip-on lights may not be (**Figure: Clip-On Light**). However, these lights can work in the gym when filming training content, at home when performing video consultation calls, or when administering virtual coaching sessions to the client.

Some free or low-budget solutions to lighting might be as simple as opting to film content in a(n):

- Brighter location on sunny days
- Office that has a large amount of natural daylight
- Room utilizing a lamp fixture with a cool white bulb installed and the lampshade removed

While these options may be less-optimal, they can help the fitness professional create higher-quality content without investing too much into equipment.



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FigureRing Light



Vetertravel/Shutterstock.com

FigureClip-On Light

Camera

One struggle that the fitness professional may encounter when building a virtual coaching business is that it lacks the same face-to-face contact as in-person fitness coaching. With over 50% of human communication conveyed through body language (Park & Park, 2018), being able to film quality workout videos, take photos for demonstrations, perform video consultation calls, and have regular video check-ins may provide higher client engagement.

The considerations when looking for a camera include the following:

- Megapixel (MP) rating of the camera
- Video recording resolution
- Budget

Megapixel Rating

While a bit of data is the smallest portion of data that a computer processes, a **pixel** is the smallest portion of an image. They are tiny little dots or squares of color that, when put together, make up an image. A digital image may look similar to a regular image but when zoomed in close enough, a digital image is made up of millions of these tiny squares. A **megapixel** is 1 million pixels, and the more megapixels in an image, the more detail and clarity an image will have. Notice that in **Figure: Close-Up of Pixels**, each little dot is actually a little square of color.



Pikepicture/Shutterstock.com

FigureClose-Up of Pixels

The megapixel rating of a camera will have a determinant effect on the quality of the photos being taken. While there is no set recommendation for how many megapixels are needed for creating content, generally speaking, a camera with a higher number of megapixels will take higher-quality photos than a camera with a lower megapixel rating. The fitness professional should consider purchasing a camera with the highest megapixel rating that is within their budget.

Pixel

The smallest portion of an image. They are tiny little dots or squares of color that, when put together, make up an image.

Megapixel

1 million pixels put together, a measure of image resolution or graphic resolution.

Video Resolution

Video resolution is another representation of image quality. The higher the video resolution, the higher the video quality. Much like an image is made from pixels, video resolution relates to how many pixels the video image has both in horizontal count and vertical count. 720p, 1080p, and 4K ultra HD are the most common resolutions for video content. Most modern digital cameras will have at minimum 1080p full HD video resolution capabilities and that is really all that would be needed for filming video content, conducting virtual coaching sessions, or performing video consultation calls. **Table: Comparing Different Video Resolutions** explains these different video resolutions.

Comparing Different Video Resolutions

Resolution	Horizontal Pixels x Vertical Pixels	Total Number of Pixels	Other names
720p	1,280 x 720 pixels	921,600 pixels	High-Definition
1080p	1,920 x 1,080 pixels	2,073,600 pixels	Full High-Definition
4K	3,840 x 2,160 pixels	8,294,400 pixels	Ultra High-Definition

Budget and Peripherals

Budget is the final consideration when purchasing a camera for a virtual coaching business. There are a few camera styles to look at and the fitness professional should do their own research to determine what is best for their needs and what is available within their price range.

- **DSLR camera:** Professional quality photos, typically films video in HD, higher price point, and may need to buy attachments
- **Hand-held digital camera:** Lower-quality but hand-held, no additional lenses needed, and reasonably priced
- **Smartphone camera**: Newer phones will have all of the features needed to produce high-quality video and images

Getting Technical

The iPhone has had HD 1,080 p recording capabilities since the iPhone 5 was released in late 2012, and Samsung, with their S4 in spring of 2013, also has 1,080 p HD recording capabilities. Earlier phones still had a respectable 8-megapixel rating and 13-megapixel rating for the S4. However, these earlier smartphones may not have some functionality like changing specific settings or auto-focus that newer smartphones may have. While there is no standard recommendation for megapixel requirements for high-quality photos, most smartphones will have a 10+ megapixel camera. If the fitness professional has a relatively new smartphone, they will be able to shoot, record, edit, and upload much of their content right from the smartphone itself.

One upgrade to consider when using a camera for recording content or even taking photographs is to purchase a quality tripod. Options vary on this due to the variety of brands and models of cameras and smartphones. It will be up to the fitness professional to research what will work best with their camera or smartphone.

In addition to capturing quality content, most laptops will have a built-in webcam. Most newer computers will have a webcam with at least 720p HD resolution for filming. While 720p is considered a lower-quality resolution for filming, a camera with a 720p resolution is still usable to get started with filming content for a virtual coaching business. There are also a multitude of USB webcam options available that will plug into the fitness professional’s computer system and may have higher resolution ratings. A webcam is a lower-budget option that can easily improve content quality without a large initial expense.

Audio

When filming or recording high-quality content for a virtual coaching business, fitness professionals should ensure that the sound quality of the content being produced is considered. Poor audio quality or videos with excessive background noise may lead to the client retaining less of the information and impair learning and communication (Klatte et al., 2013). This may lead to added struggles and create a discord between the client and the fitness professional. Purchasing a microphone can help create a quality audio feed when recording content such as podcasts, training videos, and voice-overs, and when performing virtual coaching sessions.

There are two main types of microphones to consider:

- Dynamic Microphones
- Condenser Microphones

Dynamic microphones are the types of microphones one might see being used by a singer or comedian performing on stage (**Figure: Dynamic Microphone**). These microphones tend to do better than condenser microphones in environments that have a higher level of background noises. That is why dynamic microphones are used in concerts, during stand-up shows, or interviewing somebody on the street. They contain a coil and magnet inside that are sensitive to sound vibrations. Dynamic microphones are also more rugged than condenser microphones, so, if the microphone gets dropped, there may be less worry of damage and having to replace an expensive peripheral. These microphones are relatively cheap compared to condenser microphones and require no extra equipment. They can sit at a desk in the quiet to be used for podcast recordings or used in a noisy environment such as filming workout videos.



Soonios Pro/Stuttershock.com

FigureDynamic Microphone



Krakenimages.com/Shutterstock.com

FigureCondenser Microphone

Condenser microphones work slightly differently than their dynamic counterpart (**Figure: Condenser Microphone**). Condenser microphones do not contain a coil and magnet and instead use capacitor plates, which are more sensitive to the smaller sound vibrations than a dynamic microphone. This can help pick up the subtle differences in a voice and may be better for recording a podcast in a quiet area. Condenser microphones are also not as durable as a dynamic microphone due to their construction using capacitor plates. Condenser microphones may also need an external power source or internal battery to operate. Depending on the type of microphone purchased may mean hidden equipment costs such as extra audio equipment. However, condenser microphones that are powered by battery or USB plug-in are becoming more available making these microphones a viable option for fitness professionals without added costs.

The next consideration for microphones is whether to choose a USB microphone or an XLR microphone. Simply put, a USB microphone will plug right into the computer, an XLR microphone will need an audio interface between the computer and the microphone (**Figure: USB Microphone**). A USB microphone is simple and considered *plug and play*, with no additional software, drivers, or expensive audio interface systems. XLR microphones can become more expensive due to both initial purchase of the microphone and required add-ons needed to operate (**Figure: XLR Microphone)**. However, XLR microphones (dynamic or condenser) will produce a higher-quality sound than a USB microphone.



Random Photo Gallery/Shutterstock.com

FigureUSB Microphone



Sockagphoto/Shutterstock.com

FigureXLR Microphone

One final option is to simply use a smartphone with a voice recording app. Fitness professionals should be mindful that this option will not produce the highest-quality audio recordings. Most smartphones come with a voice recording app that may have adjustable settings. There are also third-party apps that may increase recording sound quality. The fitness professional should research what may be best for this option.

If the fitness professional is also filming live workouts or doing large group instruction over video conference software, purchasing an additional wearable microphone may be needed and can improve audio quality and engagement with the clients (**Figure: Wearable Microphone**).

The Table: Dynamic Microphone Versus Condenser Microphone compares and contrasts the pros and cons of microphones. The fitness professional can use this as a guide before purchasing a microphone.

Di Studio/Shutterstock.com

FigureWearable Microphone
Dynamic Microphone Versus Condenser Microphone

	Dynamic Microphone	Condenser Microphone

Budget: When considering budget, how the microphone connects is important. USB will be plug and play, whereas XLR microphones need an interface to record.	A dynamic microphone will typically be cheaper than a condenser microphone. Using a dynamic microphone with a USB connection will be budget friendly and plug and play with no audio interface needed.	Condenser microphones are generally more expensive than a dynamic microphone. A condenser microphone that uses an XLR connection may be expensive due to initial costs because of the added audio interface that is required for XLR microphones.
Environment Considerations: Where is the audio being recorded?	A dynamic microphone will be better suited for noisy environments.	A condenser microphone will be better suited for quiet environments.
Durability: Will the fitness professional be travelling around?	A dynamic microphone is more rugged than a condenser microphone and may be better suited to the more-mobile fitness professional.	A condenser is well-suited for the less-mobile fitness professional.

Fitness Programming Software Options

There are a multitude of fitness, nutrition, and workout apps available for both the client and the fitness professional to utilize in reaching their health and fitness goals. Studies have shown that people may have greater adherence to their health goals when using mobile health apps to self-monitor and stay accountable (Voth et al., 2016). While no single app, no matter how good it is, can replace a qualified and effective fitness professional, using these apps may increase a client's engagement and adherence to their program.



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Nutrition Apps

While tracking nutritional intake has numerous benefits, many of the new apps do much more than simply count calories. This can be especially important for those who are trying to lose weight, build muscle, or improve athletic performance. Nutritional apps are beneficial for the following reasons:

- Meal planning
- Understanding and counting calories
- Reading and understanding food labels
- Tracking macronutrient intake (protein, carbohydrates, and fat)
- Tracking nutritional content (vitamins, minerals, and fiber)
- Making healthier food choices
- Fueling workouts
- Finding recipes
- Forming positive eating habits

Coach's Corner

The fitness professional should remember that when recommending that clients track or monitor meals away from home or at restaurants, there can be an 8 to 18% difference in what is indicated on the label versus the actual calories being eaten, and, in some cases, there can be up to 200% more calories. (Urban et al., 2010) All it takes is a heavy-handed chef to add a little extra oil or sauce to a dish to

increase the calorie value of a meal. It may be a good idea to have the client over-estimate meals out by 30% and also account for an additional 100 to 200 calories worth of fat from oil or butters being used. The fitness professional and client should work together to help estimate the calorie content of food.

Fitness Apps

Fitness apps are downloadable applications that can be used on nearly any mobile device and are designed to help users obtain certain fitness goals. A fitness app is an inexpensive and convenient way for fitness enthusiasts to take their workouts with them and include a variety of exercise formats, including high-intensity interval training (HIIT), strength training, yoga, and Pilates.

The benefits of fitness apps include the following:

- Access to exercise routines anytime or anywhere with a computer, tablet, or smartphone
- Personalized fitness programs based on the user's goals and exercise preferences
- Access to exercise programs that do not require equipment
- Tracking completed workouts and time spent exercising
- Daily/weekly/monthly reminders to complete workout routines
- Simulation of one-on-one or group training
- Tracking steps taken each day
- Tracking calories expended from physical activity
- Tracking fitness achievements like miles run or weight lifted
- Instructing proper exercise technique

There are several fitness apps on the market today and many have free trial periods. Individuals can test several applications to determine which app best fits their needs. Remember that individuals with chronic disease may need medical clearance before starting a new exercise routine. In addition, performing exercise routines without supervision can be dangerous, especially for those new to exercise.

NASM EDGE App

NASM has their own app for fitness professionals called NASM EDGE (**Figure: NASM EDGE 1 & 2**).

The NASM EDGE app has the following features:

- **Exercise Video Library:** Assess and share the hundreds of NASM expert-approved exercises.
- **Calorie and Macronutrient Calculator**: Give clients confidence with nutrition advice that aligns with their goals.

- **Workout Builder**: Create customized workouts or use ready-to-go training sessions designed by NASM experts.
- **Integrated Client Dashboard**: Keep track of clients and monitor their real-time performance, compare dynamic assessments, and send clients workouts.
- **Assessment Wizard**: Remove any uncertainty out of client assessments while performing them quickly and effectively.

The app also has a built-in Certified Personal Trainer (CPT) prep tool for those studying for their NASM-CPT examination. The prep tool includes over 600 practice questions, over 500 flash cards, and an unlimited number of attempts on the NASM practice exam.



FigureNASM EDGE 1



FigureNASM EDGE 2

Wearables

Wearable fitness technology has been around as early as 1981, when Olympic medalists were using wearable heart rate monitors. Commercial gyms started seeing wireless heart rate monitors in the early 2000s being integrated into group workout classes. Since 2010, when Fitbit released the first clip-on pedometer, technology has gone from basic step counting to tracking heart rate, sleep, **energy expenditure**, and even GPS tracking.

Clients can track virtually every meaningful data point about themselves, which can be helpful but can also provide additional confusion in an already overwhelming abundance of information. Keep in mind, the current body of evidence is mixed regarding the weight-loss and health benefits from wearable fitness technology (Jo et al., 2019). While the direct benefits for the client may still be in question from a research perspective, there are valuable metrics that can be tracked such as heart rate, sleep, and energy expenditure. The fitness professional can use this information to help customize and evaluate exercise programs. When considering what is best for a client, note that some of the metrics being tracked by wearable technology will be more accurate than others.

Heart Rate



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There are multiple studies showing that heart rate monitoring will be most accurate at rest or at moderate activity levels, with accuracy decreasing as exercise intensity increases (Khushhal et al., 2017). If a client is doing higher-intensity or long-duration cardiovascular training such as cycling or running, a better option to recommend is a fitness tracker that comes with a wearable heart rate chest strap (Reddy et al., 2018). If a client is looking for general heart rate monitoring, a wristband style fitness tracker will suffice.

Energy Expenditure

Many fitness trackers will estimate **total daily energy expenditure (TDEE)** and provide the end user with a calorie burn count at the end of each day. How accurate is this measure of TDEE? Studies show that most wearable fitness trackers have a range of accuracy and, much like heart rate, as exercise intensity increases, measured energy expenditure accuracy may decrease (Kendall et al., 2019). When compared to research-grade fitness trackers, resting calorie expenditure may be close. However, these

devices tend to overestimate time spent in higher-intensity activities, which leads to higher-than-actual energy expenditures being reported by the wearables (Feehan et al., 2018). This simply means that current wearables may do well at calculating energy expenditure at rest, but the level of accuracy decreases as exercise intensity increases. One study showed a 20 to 50% difference in actual calories expended versus what the wearables listed when recording energy expenditure during resistance training (Bai et al., 2016). However, it is important to note that the quality of the device's electronics and algorithms have a fundamental influence on the accuracy of the device. In other words, some wearables are more accurate than others. Due to the high degree of inconsistency across wearable fitness trackers, fitness professionals should use caution when relying on this information.

Energy expenditure

The number of calories burned over a specified period of time at a given intensity of movement.

Total daily energy expenditure (TDEE)

The number of calories a person burns over a 24-hr period.

Step Counters/Pedometers

Most step counters are fairly accurate when it comes to measuring stride length and accurately representing step counts, with an average error of 3 to 6% on some brands (Feehan et al., 2018). There are also multiple studies that show an improvement in all-cause mortality that is associated with higher step counts (Dwyer et al., 2015). In addition to the health benefits of a higher step count, monitoring step counts can help a fitness professional monitor for a subconscious decrease in **Non-Exercise Activity Thermogenesis (NEAT)** that is often associated with weight-loss-based goals among clients (Trexler et al., 2014).



Kaspars Grinvalds/Shutterstock.com

NEAT includes all movements and activities of daily living that are not structured exercise such as fidgeting, performing household chores, grooming, and pacing. If a client starts out at a 10,000-per-day step count average at the start of a weight-loss program and through their program ends up subconsciously reducing step counts to 7,000 per day, there may be a notable difference in TDEE. By monitoring step counts, clients can be encouraged to move more and avoid this downward metabolic adaptation that occurs while working to achieve a fat-loss-related goal. This is where wearable fitness technology excels in terms of accuracy, alongside being the most usable data for the fitness professional.

Coach’s Corner

The fitness professional should always try to meet the client where they are at when it comes to step counts. Assigning a 10,000-per-day step count to a client who is only achieving 2,000 steps per day may feel overwhelming to the client. Try to encourage the client to move more where they can while helping them understand NEAT can impact daily energy expenditure. Set small goals, even just 5 to 10 minutes of extra movement a day may contribute greatly to overall step counts over a week.

Sleep

One very popular function of many fitness trackers that has evolved over time is the ability to monitor sleep. Sleep impacts appetite (Brondel et al., 2010) and hunger and can affect the client's rate of recovery from their training sessions. Most fitness trackers have sleep monitors that are shown to be relatively accurate when compared to self-reported sleep time (Lee et al., 2018).



Andrey_Popov/Shutterstock.com

While the sleep trackers on wearable fitness technology may not be on par with the technology used to monitor sleep in clinical research studies, for the client and the fitness professional, the accuracy of sleep tracking in wearable fitness technology will suffice in assessing sleep hygiene. Monitoring a client's sleeping patterns may help the client to improve sleep hygiene and overall sleep quality.

Food for Thought

6 hours and 40 minutes is the average amount of sleep an American gets (Sleep Foundation, 2014).

Non-exercise activity thermogenesis (NEAT)

The measurement of energy expenditure from all other movements outside of intentional exercise activity.

Integrating Apps and Wearables Into Business

Whether using the NASM Edge App, third-party client management software, or any of the other apps listed, it is important for the fitness professional to understand how to best use these integrations when operating a business. Remember that apps and software programs have their advantages and disadvantages and there is no one correct approach for building a virtual coaching business. Budget, business scaling, client population, and ease of client use all play a role in determining which programs and apps to use. For example, when getting started, a fitness professional may choose to simply use spreadsheets for workouts and nutrition tracking. This simple system can be a low-cost solution for the fitness professional and can help their clients build effective health habits. Over time, integrating wearables such as a pedometer or activity tracker can help the client more accurately track their physical activity and will add additional value.

If a fitness professional is working with a higher client load or scaling a business and building a team of coaches, using an all-in-one client management software program may lead to a more scalable and productive approach. All-in-one client management software systems may incorporate the functions a fitness professional may need:

- Billing and invoicing systems
- Client reminder and check-in systems
- Brand customization
- Nutrition tracking
- Syncing with wearable fitness to track activity
- Workout videos and exercise programming tools

Even if the fitness professional is not operating with a high volume of clients, many of the benefits that client management software systems offer will help a small-scale online business run smoothly and productively.

The software and app options are major considerations for creating the foundations of client management and a professionally delivered coaching experience in a virtual coaching business. Each client management software system has their own unique features; therefore, it is important to conduct further research and consider all aspects that will be needed for operating a virtual coaching business.

While these options and recommendations have the fitness professional in mind, it is also important to consider the client when using these different approaches. The less-complicated the system being used, the lower the barrier of effort it will be for the client. This lower barrier of effort for coaching interaction

means a client can focus on the coaching itself rather than learning a complicated system. For some, filling out spreadsheets may be an effortless portion of their check-ins, but for others it may be a barrier of effort that causes additional struggles in adherence to the coaching program. Using a simple system such as a client management software program may allow the fitness professional to reduce time spent performing other tasks and give their clients a higher level of service. It also may reduce these critical barriers of effort for the client.

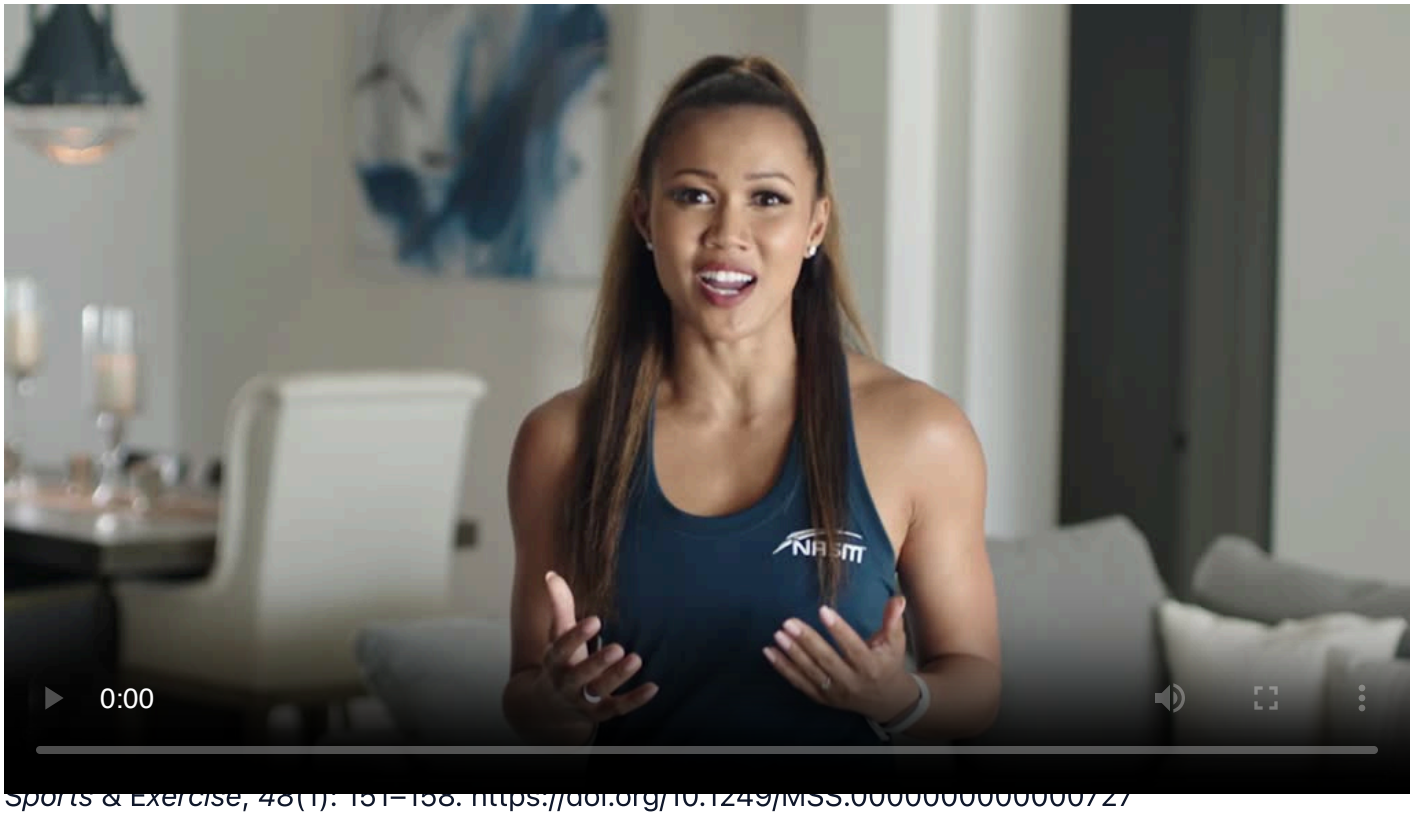
Summary

This chapter outlined what hardware, software, and peripherals are needed to deliver a highly professional and individualized coaching experience to the client.

Fitness professionals who offer virtual coaching must become knowledgeable about computer hardware and software systems. These systems include computer processing and storage, operating software, lighting, and cameras.

Fitness professionals must also acquire basic knowledge of fitness programming software, including nutrition and fitness apps and wearable technology.

Technology considerations when building a virtual coaching business are an important part of delivering a high-quality and personalized service to a client. With technology becoming more prevalent within the fitness industry, the fitness professional who can adapt and grow with these ever-changing technologies can build a successful and sustainable virtual coaching business. Please refer to the video below for a summary of the topics discussed within this chapter.



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NASM VCS Lecture CH3

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